

Mass number, atomic number and isotopes

Atomic structure
Paper 1 · Combined and Separate
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	State the overall charge of an atom.	Zero	☐
2	Explain why atoms have an overall neutral charge.	The number of protons equals the number of electrons.	☐
3	State what is the same about all atoms of the same element.	The number of protons	☐
4	State what the atomic number of an atom tells you.	The number of protons	☐
5 ★	State what the mass number of an atom tells you.	The total number of protons and neutrons	☐
6 ★	Define isotopes.	Atoms of the same element with different numbers of neutrons.	☐
7	Which differs between two isotopes of an element: mass number or atomic number?	The mass number	☐
8	Describe how ions form from atoms.	Atoms lose or gain electrons.	☐
9	An atom of carbon is written $^{14}_6\text{C}$. State what the numbers 14 and 6 represent.	14: the mass number (protons + neutrons). 6: the atomic number (protons).	☐
10 ★	State how to calculate the number of neutrons in an atom.	Number of neutrons = mass number – atomic number	☐
11 ★	State the type of ion formed when an atom loses one or more electrons.	A positive ion	☐